

Project Initialization and Planning Phase

Date	12 July 2026
Project Title	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop a data analytics solution that analyzes electricity consumption across Indian states using Microsoft SQL Server, Tableau Public, and Flask to generate interactive dashboards and meaningful insights.
Scope	The project covers data collection, SQL analysis, Tableau dashboard creation, story visualization, and Flask-based web deployment for electricity consumption analysis.
Problem Statement	
Description	Electricity consumption data is distributed across multiple datasets, making it difficult to identify trends, compare states, and evaluate consumption changes over time.
Impact	This leads to slower decision-making, inefficient reporting, and difficulty in planning energy distribution.

Proposed Solution	
Approach	Store and process electricity consumption data using Microsoft SQL Server, analyze it with SQL queries, visualize insights in Tableau dashboards, and deploy them through a Flask web application.
Key Features	• Interactive Tableau Dashboards • Tableau Story • State-wise Analysis • Region-wise Analysis • Monthly & Quarterly Trends • Lockdown Impact Analysis • Flask Web Application • Responsive UI

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 / AMD Ryzen 5 (or above), Quad Core Processor
Memory	RAM specifications	8 GB RAM
Storage	Disk space for project files, database, and dashboards	256 GB SSD (Minimum)
Software		
Frameworks	Web Framework	Flask
Libraries	Additional libraries	pandas, numpy, pyodbc, SQLAlchemy

Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	India Electricity Consumption Dataset (2019–2020), CSV Format